Management for Urgent Femur Fixation in a Post-Lung Transplant Patient on Apixaban with Chronic Kidney Disease and Distorted Neuraxial Anatomy

Venkatsai Bellala, ScB [1] Danny Bui, MD [2]

[1]Warren Alpert Medical School, Providence, RI, USA [2]Brown University Health, Providence, RI, USA

Introduction

- **Timely hip fracture surgery (within 72 hours)** is critical in reducing *1-year mortality* in elderly patients, but optimizing **complex comorbidities** remains a key challenge
- We present a case involving an elderly patient with **hematologic, musculoskeletal, pulmonary, and renal** issues that affected anesthetic and perioperative management

- **Apixaban (Eliquis)** is an oral direct Factor Xa inhibitor used for **AFib**, **DVT/PE**, and **thromboprophylaxis**[1]
  - Carries a risk of **spinal hematoma** with **neuraxial anesthesia** (spinal, epidural, CSE) due to bleeding in non-compressible spaces[2]
  - **ASRA guidelines** define safe timing for neuraxial anesthesia in anticoagulated patients[3]

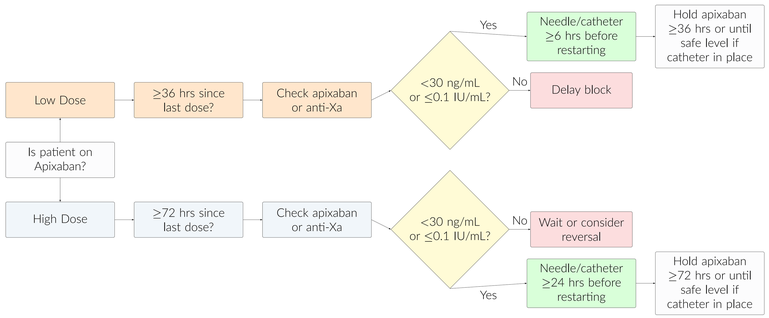

Figure 1. Flowchart of ASRA guidelines for neuraxial anesthesia in patients receiving apixaban[3]

- Further considerations apply for neuraxial anesthesia in patients receiving apixaban:[3]
  - **Routine coag tests (PT/aPTT)** are unreliable for DOAC effect
  - **Apixaban level ≤ 0.1 IU/mL** recommended before procedure (therapeutic >0.2 IU/mL)
  - Assess **respiratory status** (e.g., end-stage lung disease) during sedation planning
  - Monitor for **neurological changes** post-procedure

Impact of Chronic Kidney Disease on Apixaban Clearance

- **27% of apixaban** is renally cleared; clearance ↓ with worsening CKD (AUC ≤44%).[4]
  - Hemodialysis removes minimal apixaban (4-14%) due to high protein binding; AUC is ≈36% higher in dialysis patients
- **KCentra (4F-PCC)** is used off-label to reverse apixaban in severe bleeding or surgery:
  - In 10 patients (5 on apixaban), KCentra was safe with no deaths; PT shortened post-treatment[5]
  - 25 U/kg increased peak thrombin generation by 76% in healthy volunteers; effect declined by 48h[6]
  - Ex vivo, PCC partially reversed prophylactic apixaban; rFVIIa more effective but clinical impact unclear[7]
  - Not FDA-approved for apixaban reversal; consider with close thrombosis monitoring

Table 1. KCentra (4F-PCC) vs. Andexxa (Andexanet Alfa) for Eliquis (Apixaban) Reversal

	KCentra (4F-PCC)[6]	Andexxa (Andexanet Alfa)[8]
Type	Four-Factor Prothrombin Complex Concentrate	Modified Factor Xa decoy protein
Mechanism	Boosts thrombin generation	Binds FXa inhibitors, restoring thrombin activity
Effectiveness	↑ Peak thrombin generation (76%)	Good hemostasis (79-82%)
Monitoring	Standard coagulation tests unreliable	↓ anti-Xa activity; monitoring imperfect
Thrombotic Risk	Prothrombotic potential	Black box warning; 27% thrombotic events
FDA Approval	Off-label for apixaban	Approved for life-threatening bleeding
Cost/Availability	More available, lower cost	Expensive, limited availability

Case Description

A 76-year-old male presents following a fall and is found to have a left hip fracture.

Past Medical History

- Chronic kidney disease
- Meningocele repair
- Non-insulin dependent diabetes
- Atrial fibrillation treated with Apixaban
- Idiopathic pulmonary fibrosis post lung transplant in 2015

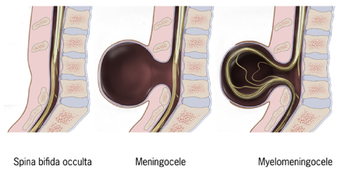

Figure 2. Types of spina bifida. Image credit: CDC.

Table 2. Timeline of Events

Date (Day)	Time	Event
02/17/25 (1)	05:00	Dyspneic after 2mg morphine, put on 5L oxygen. At home, requires 4L O_2 with activity.
02/17/25 (1)	20:00	On 15L high flow nasal cannula. Left lower lobe opacities and pulmonary edema noted. Antibiotics began for pneumonia. Diuretics administered.
02/18/25 (2)	10:20	Orthopedics consults Anesthesiology for clearance prior to booking left hip hemiarthroplasty. Patient took Eliquis night before admission. Oxygen requirements improved, but on 10L O_2. Recommended optimization and more time for Eliquis to clear before surgery.
02/19/25 (3)	07:00	On 5L O_2, Anti-Xa level 0.22 IU/mL, Orthopedics states need to operate given narrowing window to operate to avoid increase in mortality.
02/19/25 (3)	07:20	Blood bank states will possibly need >72 hours for Eliquis to clear due to chronic kidney disease. Can give KCentra and wait 20-30 minutes prior to neuraxial. Andexanet alpha is not stocked. No need for additional testing as anti-Xa level will not change.
02/19/25 (3)	09:11	KCentra 2000 units administered preoperatively.
02/19/25 (3)	09:40	Attempted spinal in lateral position due to inability to place pressure on fractured hip. Scar from L1-L5 noted. Unable to discern landmarks, ultrasound utilized without ability to recognize landmarks. Tuohy needle used with loss of resistance noted; however, no CSF aspirated with spinal needle.
02/19/25 (3)	10:14	Patient wishes to proceed with general anesthesia. Induced and intubated.
02/19/25 (3)	11:52	Procedure end. Patient extubated uneventfully.
02/19/25 (3)	13:00	Patient discharged from PACU to step down on 4L O_2.
02/20/25 (4)		Patient on room air. Only complaint are symptoms consistent with postdural puncture headache.

Discussion

Impact of Timing on Hip Fracture Surgery Outcomes

- Orthopedic consensus favors surgery within 24-48 hours of fracture for best outcomes
- Delays beyond 72 hours are often associated with worse outcomes
  - Surgery delayed >72 hrs **doubles 1-year mortality** risk (controlling for age/comorbidities)[9]
  - A meta-analysis of 257,000+ patients found >**48 hrs delay increased mortality**[10]
  - **61.9% of patients who died** within 1 year had **surgery >72 hours** after injury[11]
- In frail patients needing medical optimization, mortality may correlate more with **comorbidities** than **timing alone**

Neuraxial Anesthesia in Patients with Spinal Abnormalities

- **Prior spinal surgery or dysraphism** (e.g., meningocele repair) complicates neuraxial access due to **scar tissue, altered landmarks, and risk of tethered cord**, often requiring **imaging guidance** and alternative analgesia plans[2,12]
- A **"dry tap"** may result from a **collapsed subarachnoid space**, low CSF pressure, or technical failure; prior surgery increases this risk[13]
  - **Ligamentum flavum may be absent or distorted** post-surgery, making the **loss-of-resistance technique unreliable** for epidural placement
  - **Spinal anatomy distortion** can limit local anesthetic spread, increasing the risk of **failed or patchy block**
  - **Avoid neuraxial puncture at previous operative levels**; review **surgical records, MRI**, or consider **ultrasound guidance** to identify viable interspaces

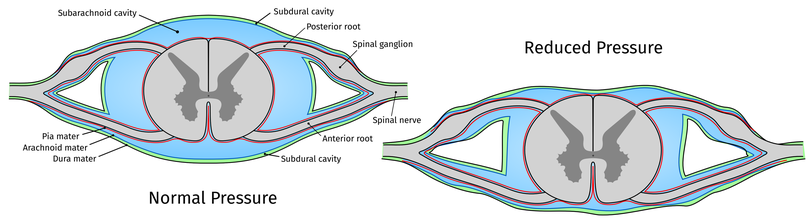

Figure 3. Illustrative comparison of the subarachnoid space under normal and reduced cerebrospinal fluid (CSF) pressure. Normal pressure maintains separation between the arachnoid and pia mater, while reduced pressure and postoperative adhesions may cause collapse and disrupt CSF flow. Adapted from Wikimedia Commons.

References

[1] A. Agrawal and B. Manna, "Apixaban," in *StatPearls*, Treasure Island (FL): StatPearls Publishing, 2025.

[2] G. Crowe and T. Drew, "Neuraxial anaesthesia in the parturient with pre-existing structural spinal pathology," *BJA Education*, vol. 24, pp. 361–370, Oct. 2024.

[3] S. L. Kopp, E. Vandermeulen, R. D. McBane, A. Perlas, L. Leffert, and T. Horlocker, "Regional anesthesia in the patient receiving antithrombotic or thrombolytic therapy: American Society of Regional Anesthesia and Pain Medicine Evidence-Based Guidelines (fifth edition)," *Regional Anesthesia & Pain Medicine*, Jan. 2025. Publisher: BMJ Publishing Group Ltd Section: Special article.

[4] S. R. Mandt, N. Thadathil, C. Klem, C. Russ, P. L. McNamee, K. Stigge, and D. Cheng, "Apixaban Use in Patients with Kidney Impairment: A Review of Pharmacokinetic, Interventional, and Observational Study Data," *American Journal of Cardiovascular Drugs*, vol. 24, no. 5, pp. 603–624, 2024.

[5] K. Gundabolu, B. Trevarrow, and S. A. Koepsell, "Efficacy and Safety of Four Factor Prothrombin Complex Concentrates As a Reversal Agent for Oral Xa Inhibitors," *Blood*, vol. 126, p. 4668, Dec. 2015.

[6] S. Nagalla, L. Thomson, Y. Oppong, B. Bachman, I. Chervoneva, and W. Kraft, "Reversibility of Apixaban Anticoagulation with a Four-Factor Prothrombin Complex Concentrate in Healthy Volunteers," *Clinical and Translational Science*, vol. 9, no. 3, pp. 176–180, 2016. _eprint: https://ascpt.onlinelibrary.wiley.com/doi/pdf/10.1111/cts.12398.

[7] K. Schmidt, K. Krüger, E. Langer, M. Schmutzler, E. Johnen, K. D. Wernecke, C. von Heymann, and M. K. Körber, "Reversal of apixaban induced alterations in haemostasis by different coagulation factor concentrates in patients after hip or knee replacement surgery," *Blood Transfusion*, vol. 17, pp. 157–162, Mar. 2019.

[8] P. G. Bradshaw, S. P. Keegan, M. E. Droege, N. J. H. Dykes, N. E. Ernst, M. J. Foertsch, A. T. Makley, E. W. Mueller, C. D. Philpott, V. Srinivasan, J. B. Winter, M. D. Goodman, and C. A. Droege, "Reversal of apixaban and rivaroxaban with andexanet alfa prior to invasive or surgical procedures," *Pharmacotherapy: The Journal of Human Pharmacology and Drug Therapy*, vol. 42, no. 10, pp. 780–791, 2022. _eprint: https://accpjournals.onlinelibrary.wiley.com/doi/pdf/10.1002/phar.2727.

[9] J. D. Zuckerman, M. L. Skovron, K. J. Koval, G. Aharonoff, and V. H. Frankel, "Postoperative complications and mortality associated with operative delay in older patients who have a fracture of the hip," *The Journal of Bone and Joint Surgery. American Volume*, vol. 77, pp. 1551–1556, Oct. 1995.

[10] N. Simunovic, P. Devereaux, and M. Bhandari, "Surgery for hip fractures: Does surgical delay affect outcomes?," *Indian Journal of Orthopaedics*, vol. 45, no. 1, pp. 27–32, 2011.

[11] A. Lari, A. Haidar, Y. AlRumaidhi, M. Awad, and O. AlMutairi, "Predictors of mortality and length of stay after hip fractures – A multicenter retrospective analysis," *Journal of Clinical Orthopaedics and Trauma*, vol. 28, p. 101853, Apr. 2022.

[12] V. Gupta and G. Arora, "Emergency Cesarean Section under Spinal Anesthesia in an Operated Patient of Meningocele during Her Infancy: A Case Report," *Journal of Anaesthesia and Critical Care Case Reports*, vol. 4, no. 2, pp. 24–26, 2018.

[13] K. Ramachandran and N. Ponnusamy, "Dry tap and spinal anesthesia," *Canadian Journal of Anesthesia/Journal canadien d'anesthésie*, vol. 52, pp. 1104–1105, Dec. 2005.

bellala@brown.edu • DBui@brownhealth.org